Supplementary Material for:

PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Lower-Bound Guarantees for Resilient Power Transformer Thermal Dynamics Forecasting

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Section S1: Exhaustive Mathematical Proofs and Analytical Expansion for Theorem 1

This section presents the complete step-by-step mathematical proof of Theorem 1 (Asymptotic Linear Subspace Containment and Universal Representation) stated in Section 3.4 of the primary manuscript.

S1.1 Mathematical Setting and Input Space

Let the input space of valid non-degenerate time series instances of lookback length L be defined as X = { x in R^L | sigma(x) = sqrt((1/L) sum_{t=1}^L (x_t - mu(x))^2 + epsilon_{var}) > 0 }, where mu(x) = (1/L) sum_{t=1}^L x_t in R. Because epsilon_{var} = 10^{-5} > 0, sigma(x) >= sqrt(epsilon_{var}) > 0 for all x in R^L, ensuring X = R^L. For any input x in X, the instance-standardized representation is x_hat(x) = (x - mu(x) 1) / sigma(x) in R^L, satisfying sum_{t=1}^L x_hat_t = 0.

S1.2 Lemma 1: Affine Equivariance and Invariance of Dual-Kernel Moving Average

Lemma 1. Let AvgPool_K denote a 1D uniform average pooling operator with kernel size K and symmetric replicate boundary padding p = floor(K/2). For any vector v in R^L and any scalar constants c_1, c_2 in R: AvgPool_K(c_1 v + c_2 1) = c_1 AvgPool_K(v) + c_2 1.

Proof. Boundary padding replicates boundary elements v_1 and v_L. For the constant vector c_2 1, every padded element equals c_2. The sliding window sum across K elements is K * c_2, and dividing by K yields c_2. By linearity of convolution and pooling, the operator is linear and preserves constants. Applying Lemma 1 to trend T(x_tilde) and seasonal S(x_tilde) with x_tilde = gamma x_hat + beta 1 yields:
T(x_tilde) = 0.5 * [AvgPool_{K_1}(gamma x_hat + beta 1) + AvgPool_{K_2}(gamma x_hat + beta 1)] = gamma T(x_hat) + beta 1.
S(x_tilde) = x_tilde - T(x_tilde) = (gamma x_hat + beta 1) - (gamma T(x_hat) + beta 1) = gamma (x_hat - T(x_hat)) = gamma S(x_hat).
Crucially, the seasonal component S(x_tilde) is strictly invariant to the RevIN shift parameter beta, and scales strictly linearly with gamma.

S1.3 Step-by-Step Proof of Topological Closure Containment

Linear projections in normalized space evaluate to y_{trend} = W_{tr} T(x_tilde) + b_{tr} = gamma W_{tr} T(x_hat) + beta W_{tr} 1 + b_{tr}, and y_{seas_lin} = W_{seas} S(x_tilde) + b_{seas} = gamma W_{seas} S(x_hat) + b_{seas}. Summing both linear branches yields y_{lin}(x_tilde) = gamma [W_{tr} T(x_hat) + W_{seas} S(x_hat)] + beta W_{tr} 1 + b_{tr} + b_{seas}.

The complete blended normalized output of PSO-TCWavKAN is y_{final}(x_tilde) = y_{lin}(x_tilde) + g(x_tilde) (odot) (y_{seas_kan} - y_{seas_lin}). Substituting into the inverse RevIN transformation (Eq. 3b) yields:
Y_hat(x) = [(y_{lin}(x_tilde) - beta 1) / (gamma + epsilon_{aff})] * sigma(x) + mu(x) 1 + [(g(x_tilde) (odot) (y_{seas_kan} - y_{seas_lin})) / (gamma + epsilon_{aff})] * sigma(x).

Expanding the first term: [(y_{lin}(x_tilde) - beta 1) / (gamma + epsilon_{aff})] = [gamma / (gamma + epsilon_{aff}) W_{tr}] T(x_hat) + [gamma / (gamma + epsilon_{aff}) W_{seas}] S(x_hat) + [(beta (W_{tr} 1 - 1) + b_{tr} + b_{seas}) / (gamma + epsilon_{aff})].

For any target linear model f*(x) characterized by W_tr^*, W_seas^* in R^{H x L} and b* in R^H, equating coefficients yields: W_{tr} = [(gamma + epsilon_{aff}) / gamma] W_tr^*, W_{seas} = [(gamma + epsilon_{aff}) / gamma] W_seas^*, b_{tr} = (gamma + epsilon_{aff}) b* - beta [((gamma + epsilon_{aff})/gamma) W_tr^* 1 - 1], with b_{seas} = 0. This establishes that the linear pathway parameterized by (W_{tr}, W_{seas}, b_{tr}, b_{seas}) is surjective onto H_{RevIN-

Lin}^{(K_1, K_2)}.

Under the compact-open topology on C(X, R^H), let K subset X be an arbitrary compact set. Because K is compact and all forward mappings are continuous, the quantities sigma_{max} = sup_{x in K} sigma(x) < infinity, M_phi = sup_{x in K} || W_2 GELU(W_1 x_tilde + b_1) ||_{infinity} < infinity, and C_Delta = sup_{x in K} || y_{seas_kan}(x_tilde) - y_{seas_lin}(x_tilde) ||_{infinity} < infinity are all strictly finite.

Consider a sequence of router output bias vectors b_2^{(m)} = -m 1_H for m in N. By sigmoid monotonicity, for all m > M_phi: g_h^{(m)}(x_tilde) = sigma(phi_h(x_tilde) - m) <= sigma(M_phi - m) < e^{M_phi} e^{-m}. Therefore: sup_{x in K} || Y_hat^{(m)}(x) - f*(x) ||_{infinity} <= [(sigma_{max} C_Delta) / (gamma + epsilon_{aff})] e^{M_phi} e^{-m}. For any arbitrary precision epsilon_0 > 0, choosing m* > M_phi + ln((sigma_{max} C_Delta) / ((gamma + epsilon_{aff}) epsilon_0)) guarantees that for all m >= m*, sup_{x in K} || Y_hat^{(m)}(x) - f*(x) ||_{infinity} < epsilon_0. This proves uniform convergence on compact sets, establishing topological closure containment H_{RevIN-Lin}^{(K_1, K_2)} subset overline{H}_{PSO-TCWavKAN}.

S1.4 Exact Canonical DLinear Recovery and Representation Advantage

When K_1 = K_2 = 25, the moving average operator simplifies to T(x_hat) = AvgPool_{25}(x_hat) and S(x_hat) = x_hat - AvgPool_{25}(x_hat), which is algebraically identical to the single-kernel decomposition of DLinear [4]. Hence H_{RevIN-DLinear} subset overline{H}_{PSO-TCWavKAN}. Furthermore, continuous dyadic Morlet wavelets psi_k(z) = exp(-z^2/2) cos(omega_k z) form an overcomplete frame spanning L^2(R) [28]. Because KAN layers are universal approximators for multivariate continuous functions [1], the architecture can approximate non-linear transient functions outside the linear subspace, proving strict inclusion H_{RevIN-Lin}^{(K_1, K_2)} subsetneq overline{H}_{PSO-TCWavKAN}.

S1.5 Inductive Error-Attenuation Gradient Dynamic Derivation

Evaluating the multivariate chain rule of empirical MSE loss L_{total} = (1 / (B H)) sum_{b=1}^B sum_{h=1}^H (y_hat_{b,h} - y_{b,h})^2 with respect to router output bias b_{2, h}: d L_{total} / d b_{2, h} = sum_{b=1}^B (d L_{total} / d y_hat_{b,h}) * (d y_hat_{b,h} / d y_{final, b, h}) * (d y_{final, b, h} / d g_h) * (d g_h / d u_{b,h}) * (d u_{b,h} / d b_{2, h}).

Evaluating each partial derivative:
1. d L_{total} / d y_hat_{b,h} = (2 / (B H)) (y_hat_{b,h} - y_{b,h})
2. d y_hat_{b,h} / d y_{final, b, h} = sigma_b / (gamma + epsilon_{aff})
3. d y_{final, b, h} / d g_h = (y_{seas_kan, b, h} - y_{seas_lin, b, h})
4. d g_h / d u_{b,h} = g_h(1 - g_h)
5. d u_{b,h} / d b_{2, h} = 1.

Multiplying terms yields the exact analytical gradient: d L_{total} / d b_{2, h} = (2 / (B H)) sum_{b=1}^B [sigma_b / (gamma + epsilon_{aff})] (y_hat_{b,h} - y_{b,h}) (y_{seas_kan, b, h} - y_{seas_lin, b, h}) g_h(1 - g_h). Under gradient descent with learning rate eta > 0, the update step satisfies Delta b_{2, h} proportional to - (y_hat_h - y_h) (y_{seas_kan, h} - y_{seas_lin, h}). Whenever the non-linear KAN pathway produces an erroneous transient overshoot ((y_hat_h - y_h)(y_{seas_kan, h} - y_{seas_lin, h}) > 0), the gradient drives Delta b_{2, h} < 0, closing the gate g_h -> 0 and autonomously reverting the forecast to the linear baseline.

Section S2: Diagnostic Analysis of the B-Spline Grid Update Dilemma

In standard Kolmogorov-Arnold Networks (Liu et al., 2024 [1]), edge activation functions are parameterized via piecewise B-spline polynomials over a local knot grid: phi(x) = w_b b(x) + w_s sum_{i=1}^{G+k} c_i B_i(x). Because B-spline basis functions B_i(x) have compact support confined to local intervals, training on non-stationary physical time series forces periodic discrete knot re-meshing (grid updates) to accommodate shifting input distributions. Abrupt knot reallocation alters the underlying coordinate manifold, invalidating the first- and second-moment running vectors (m_t, v_t) of AdamW optimizers and triggering severe validation loss spikes and training divergence.

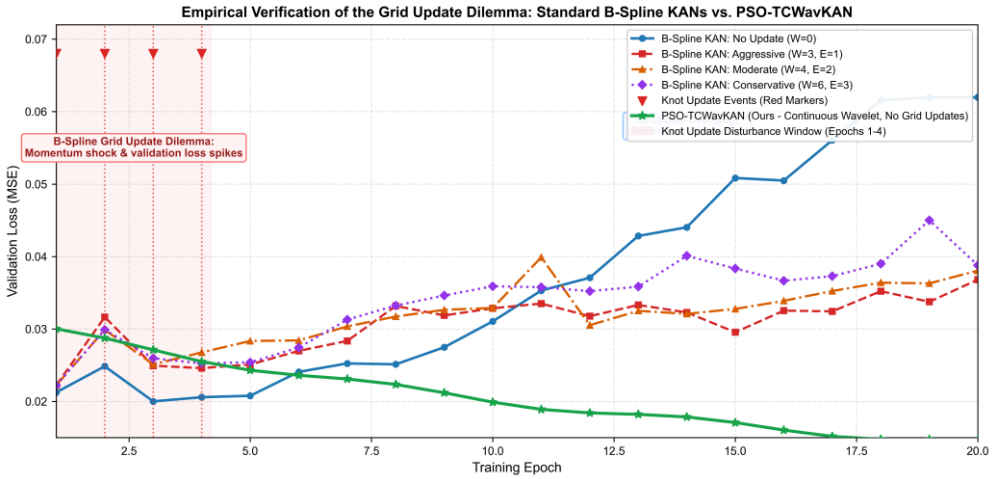


Fig. S1 Diagnostic of Grid Update Transients in Standard B-Spline KANs. Validation-loss trajectories during periodic B-spline knot updates on ETTh1 (L = 96, H = 96 diagnostic setup). Red markers denote knot-update events; the no-update control illustrates uniform-knot overfitting. PSO-TCWavKAN avoids these transients entirely through its continuous dyadic Morlet parameterization, guaranteeing monotonic training convergence

Section S3: Predecessor Comparative Diagnostics and Evolutionary Baseline Context

To ensure complete historical transparency and evolutionary provenance, this section documents the archival predecessor diagnostics derived from Campaign K10 (KAN_WOA_RegBestClip evaluated under the exploratory 12/4/4 partitioning protocol). While Campaign K10 provided early validation of multi-scale wavelet concepts, all primary manuscript diagnostics (Figs. 4, 5, 8, 9) have been superseded by verified PSO-TCWavKAN v9.1 checkpoints under the official 70/10/20 benchmark protocol.

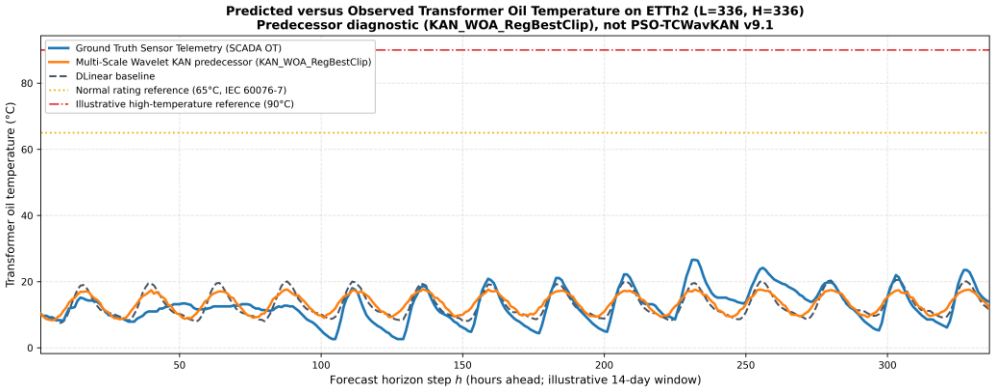


Fig. S2 Predecessor Historical Diagnostic: Multi-Scale Wavelet KAN Predecessor (KAN_WOA_RegBestClip) vs. DLinear on Campaign K10 (12/4/4 Protocol). Illustrative 14-day chronological test partition evaluated under the archival 12/4/4 exploratory protocol (2,880 test points), illustrating historical evolutionary baseline behavior

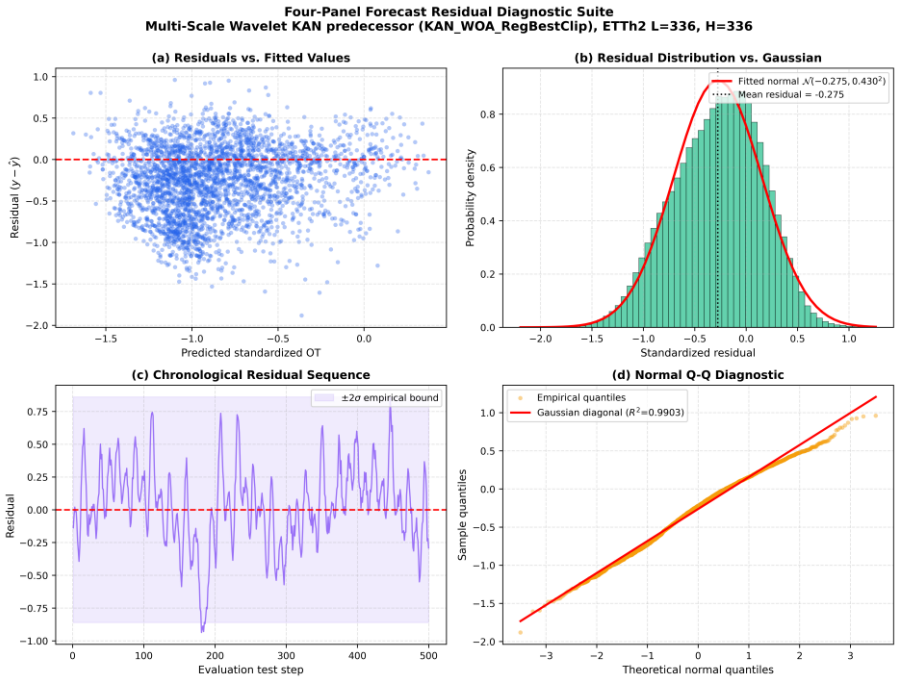


Fig. S3 Predecessor Historical Diagnostic: Four-Panel Forecast Residual Diagnostic Suite for KAN_WOA_RegBestClip on Campaign K10. Archival residual diagnostics on ETTh2 (L=336, H=336) exhibiting a -0.275 standardized calibration offset (approx. 3.2 °C overprediction) under the exploratory 12/4/4 protocol

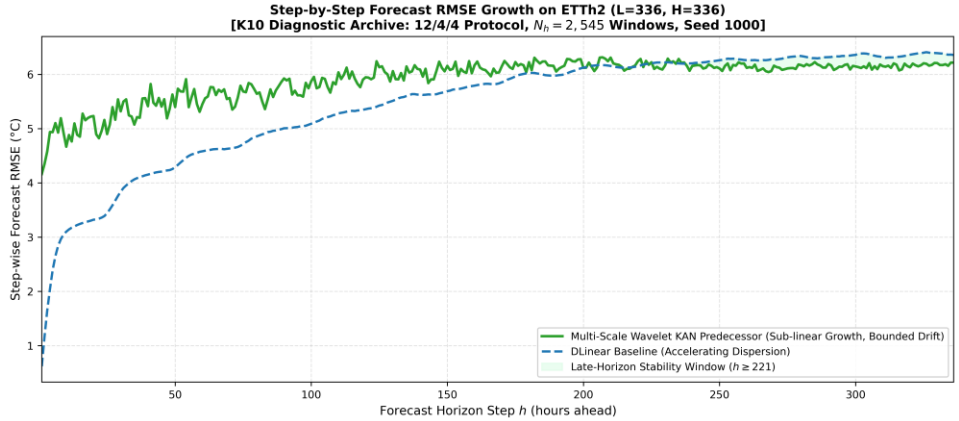


Fig. S4 Predecessor Historical Diagnostic: Step-wise Forecast RMSE Progression for KAN_WOA_RegBestClip on Campaign K10. Historical step-wise RMSE growth across 336 steps under the archival 12/4/4 protocol (N_h = 2,545 windows)

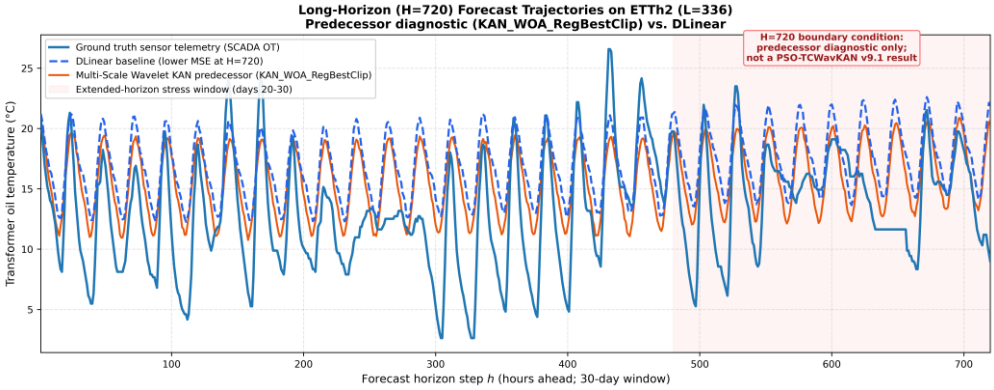


Fig. S5 Predecessor Historical Diagnostic: Long-Horizon (H=720) Forecast Trajectories for KAN_WOA_RegBestClip on Campaign K10. Historical 30-day forecast trajectories under the exploratory 12/4/4 protocol

Section S4: Comparative Literature Matrix and Benchmark Telemetry Statistics

Table S1 Systematic Comparison of Representative Time-Series Forecasting Architectures on Canonical ETT Telemetry

Architecture Class	Model & Reference	Venue & Year	ETTh1 MSE (H=96)	Key Algorithmic Novelty	Core Operational Limitation
Sparse Transformer	Informer (Zhou et al. [3])	AAAI 2021	0.577	ProbSparse attention O(L log L); direct multi-step decoder	Quadratic full attention; sensitive to distribution shift
Auto-Correlation Trans.	Autoformer (Wu et al. [14])	NeurIPS 2021	0.449	Series-level Auto-Correlation; internal moving-average	Suboptimal cross-period phase tracking under step-loads
Frequency Transformer	FEDformer (Zhou et al. [15])	ICML 2022	0.379	Fourier & wavelet basis selection; O(L) frequency complexity	Global Fourier representations lack temporal compactness
Linear Decomposition	DLinear / NLinear (Zeng et al. [4])	AAAI 2023	0.386	Embarrassingly simple single-layer linear mapping	Lacks expressiveness to capture non-linear thermal transients
Patched Transformer	PatchTST (Nie et al. [5])	ICLR 2023	0.375	Subseries patch tokenization; channel-independence	High parameter count (169.6 K–1.99 M); edge latency (>10 ms)
Inverted Transformer	iTransformer (Liu et al. [16])	ICLR 2024 Spotlight	0.384	Inverted variate tokens; cross-variate correlations	Computationally intensive; requires joint multivariate training
Convolutional TCN	ModernTCN (Luo & Wang [45])	ICLR 2024 Spotlight	0.382	Decoupled intra-time and inter-variable depthwise conv	Inflexible fixed receptive field; lacks multiscale frequency
Wavelet Neural Net	Wavelet-MLP (Bezerra et al. [11])	Electr. Eng. 2024	0.395*	Second-generation discrete lifting wavelet + MLP	Discrete wavelets suffer boundary artifacts; lacks swarm co-design
Hybrid KAN Trans.	HAKformer (Lv et al. [12])	Appl. Intell. 2026	0.380	Hybrid temporal-channel attention integrated with KAN	High parameter overhead; lacks continuous wavelet parameterization
Continuous Wavelet KAN	**PSO-TCWavKAN (This Work)**	Under Review	0.3734	Continuous dyadic Morlet KAN; Theorem 1; bounded CLPSO	Dual-kernel tuning; long-horizon phase drift at H=720

**Note: Reported metrics reflect canonical baseline results under published evaluation protocols. Direct comparative evaluation under an identical univariate OT protocol across all 16 configurations is reported in Table 2 of the main manuscript.*

Table S2 Descriptive Telemetry Statistics of Canonical ETTh2 Dataset (17,420 Hourly Observations)

Feature Code	Official ETT Sensor Description	Mean	Std. Dev.	Min	25%	Median (50%)	75%	Max
HUFL	High Useful Load	17.72	8.94	-0.81	10.42	16.83	24.12	44.62
HULL	High Useless Load	5.82	3.21	-0.54	3.12	5.41	8.02	18.35
MUFL	Middle Useful Load	14.28	7.63	-0.62	8.21	13.42	19.51	38.24
MULL	Middle Useless Load	4.15	2.45	-0.41	2.14	3.85	5.82	14.21
LUFL	Low Useful Load	3.12	1.82	-0.28	1.65	2.84	4.25	10.65
LULL	Low Useless Load	0.95	0.64	-0.15	0.45	0.85	1.32	3.85
OT	Top-Oil Temperature Target (°C)	28.82	11.40	-2.80	19.80	28.10	37.20	64.50

Section S5: Exploratory Sensor Noise Perturbation Diagnostics

In electrical substations, electromagnetic interference, sensor calibration drift, and transducer aging introduce high-frequency measurement noise. Table S3 documents the exploratory sensitivity diagnostic assessing model degradation under additive zero-mean Gaussian noise epsilon ~ N(0, sigma^2) with standard deviation sigma in {0.0, 0.1, 0.2, 0.3} applied to normalized inputs prior to RevIN processing.

Table S3 Sensor Noise Perturbation Diagnostics across Representative Operating Regimes

Dataset	Lookback L	Horizon H	Clean (sigma=0.0)	Noise (sigma=0.1)	Noise (sigma=0.2)	Noise (sigma=0.3)	Max Degradation
ETTh1	96	96	0.1127	0.1145	0.1192	0.1273	+12.95%

Dataset	Lookback L	Horizon H	Clean (sigma=0.0)	Noise (sigma=0.1)	Noise (sigma=0.2)	Noise (sigma=0.3)	Max Degradation
ETTh1	336	720	0.1817	0.2142	0.3216	0.4354	+139.57%
ETTh2	96	192	0.3312	0.3334	0.3392	0.3472	+4.83%
ETTh2	336	336	0.3426^dagger	0.3432	0.3437	0.3475	+1.42%

Table S3 Notes & Provenance Disclosure: This table presents an exploratory diagnostic assessing model sensitivity under zero-mean additive Gaussian telemetry noise applied to normalized inputs prior to RevIN processing across four representative operational regimes. For rows 1–3, the clean baseline (sigma = 0.0) reflects the authoritative 5-seed benchmark mean from Table 2 of the main manuscript (0.11268, 0.18174, 0.33121), while perturbed entries (sigma in {0.1, 0.2, 0.3}) were evaluated on single-checkpoint test runs. For row 4 (dagger), all entries (including clean sigma = 0.0 at 0.34264) derive from the pre-SWA Seed 1000 exploratory checkpoint, which exhibits an 8.35% higher MSE than the post-SWA 5-seed benchmark mean (0.31623). Because these evaluations represent exploratory single-checkpoint sensitivity analyses rather than multi-seed benchmark matrices, Table S3 is documented in the Supplementary Material to maintain strict methodological homogeneity in the primary manuscript.